CHAPTER 33

# Mensuration I: Area and Perimeter

## INTRODUCTION

In this chapter, we shall be dealing with plane figures of various shapes by finding their sides, perimeters and areas.

**Area:**

The area of any figure is the amount of surface enclosed within its boundary lines. Area is always expressed in square units.

**Units of Measuring Area**

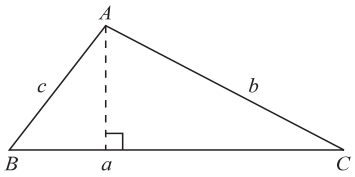
100 sq millimetres = 1 sq centimetre
100 sq centimetres = 1 sq decimetre
100 sq decimetres = 1 sq metre
100 sq metres = 1 sq decametre or arc
10,000 sq metres = 1 hectare
1,000,000 sq metres = 100 hectares
= 1 sq kilometre

### Perimeter

The perimeter of a geometrical figure is the total length of the sides enclosing the figure.

## SOME BASIC FACTS

### 1. Triangle




A *triangle* is a closed figure bounded by three sides. Here, $ABC$ is a triangle.

The sides $AB$, $BC$ and $AC$ are denoted by $c$, $a$ and $b$. respectivlely

Area of a Triangle ($A$)

(a) $A = \frac{1}{2}$ (base × height) $= \frac{1}{2}ah$

(b) $A = \sqrt{s(s-a)(s-b)(s-c)}$,

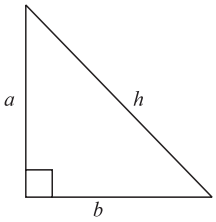
Where $s = \frac{1}{2}(a+b+c)$ or semi-perimeter of the triangle.

This formula is known as *Hero's formula*.

Perimeter ($P$) $= a + b + c = 2s$.

### 2. Right Angled Triangle

A triangle having one of its angles equal to 90° is called a *right-angled triangle*. The side opposite to the right angle is called the *hypotenuse*.




In a right angled triangle,

(Hypotenuse)$^2$ = Sum of the squares of sides

i.e., $h^2 = a^2 + b^2$.

Area ($A$) $= \frac{1}{2}$(product of the sides containing the right angle)

i.e., $A = \frac{1}{2}ab$.

**Illustration 1:** What is the area of a triangle having sides 3 m, 4 m and 5 m?

**Solution:** Let $a$ = 3 m, $b$ = 4 m, $c$ = 5 m.

Then, $s = \frac{a+b+c}{2} = \frac{3+4+5}{2} = 6$ m.

$\therefore$ Area $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{6(6-3)(6-4)(6-5)}$

$= \sqrt{6\times3\times2\times1} = \sqrt{36} = 6$ m$^2$

**Illustration 2:** Find the area of a triangle whose base is 4.6 m and height is 67 cm.

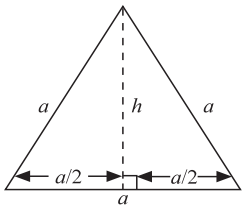
**Solution:** Area of the triangle $= \frac{1}{2}$(base × height)

$= \frac{1}{2}(4.6 \times 100 \times 67)$

$= 15410$ cm$^2$

### 3. Equilateral Triangle

A triangle whose all sides are equal is called an equilateral triangle.




Area ($A$) of an equilateral triangle

$$= \frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4}a^2$$

Permeter ($P$) of an equilateral triangle

$$= 3 \times (\text{side}) = 3a$$

Altitude ($h$) of an equilateral triangle

$$= \frac{\sqrt{3}}{2} \times (\text{side}) = \frac{\sqrt{3}}{2}a.$$

In an equilateral triangle

$$\angle A = \angle B = \angle C = 60°.$$

Area ($A$) of an equilateral triangle

$$= \frac{(\text{altitude})^2}{\sqrt{3}} = \frac{h^2}{\sqrt{3}}.$$

**Illustration 3:** Find the area of an equilateral triangle each of whose sides measures 6 cm.

**Solution:** Area of the equilateral triangle

$$= \frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4} \times 36 = 9\sqrt{3} \text{ cm}^2.$$

**Illustration 4:** Length of the side of an equilateral triangle is $\frac{4}{\sqrt{3}}$ cm. Find its height.

**Solution:** Height of the equilateral triangle

$$= \frac{\sqrt{3}}{2} \times (\text{side})$$

$$= \frac{\sqrt{3}}{2} \times \frac{4}{\sqrt{3}} = 2 \text{ cm.}$$

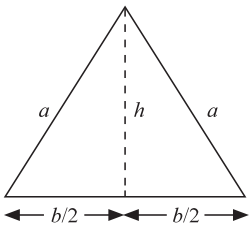
**Illustration 5:** Height of an equilateral triangle is 4 $\sqrt{3}$ cm. Find its area.

**Solution:** Area of the equilateral triangle

$$= \frac{(\text{altitude})^2}{\sqrt{3}} = \frac{4\sqrt{3}\times4\sqrt{3}}{\sqrt{3}} = 16\sqrt{3} \text{ cm}^2.$$

### 4. Isosceles Triangle

A triangle whose two sides are equal is called an *isosceles triangle*.




Area ($A$) of an isosceles triangle

$$= \frac{b}{4}\sqrt{4a^2 - b^2}$$

Perimeter ($P$) of an isosceles triangle

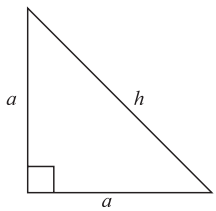
$$= (2a + b)$$

Height ($h$) of an isosceles triangle

$$= \frac{1}{2}\sqrt{4a^2 - b^2}.$$

### 5. Isosceles Right-angled Triangle

An *isosceles right-angled triangle* has two sides equal with equal sides making 90° to each other.



Hypotenuse $(h) = \sqrt{2a}$

Area $(A) = \frac{1}{2}a^2$

Perimeter $(P) = 2a + \sqrt{2a} = \sqrt{2}a(\sqrt{2}+1)$

$= h(\sqrt{2}+1)$.

If the perimeter of an isosceles triangle is $P$ and the base is $b$, then the length of the equal sides is $\left(\frac{P-b}{2}\right)$.

If the perimeter of an isosceles triangle is $P$ and the length of equal sides is $a$, then base is $(P - 2a)$.

**Illustration 6:** An isosceles right-angled triangle has two equal sides of length 6 m each. Find its area.

**Solution:** Area $= \frac{1}{2}$ (equal side)$^2 = \frac{1}{2}(6)^2 = 18$ m$^2$.

**Illustration 7:** The perimeter of an isosceles triangle is 80 cm. If the length of the equal sides is 15 cm, find the length of the base.

**Solution:** Length of the base $= P - 2a$

$= 80 - 2(15) = 50$ cm.

**Illustration 8:** The perimeter of an isosceles triangle is 42 cm. If the base is 16 cm, find the length of the equal sides.

**Solution:** The length of the equal sides $= \frac{P-b}{2}$.

$= \frac{42-16}{2} = \frac{26}{2} = 13$ cm.

**Illustration 9:** If the base of an isosceles triangle is 10 cm and the length of the equal sides is 13 cm, find its area.

**Solution:** Area of the isosceles triangle

$= \frac{b}{4}\sqrt{4a^2-b^2}$

$= \frac{10}{4}\sqrt{4\times(13)^2-(10)^2}$

$= \frac{10}{4}\sqrt{676-100} = \frac{10}{4}\times 24 = 60$ cm$^2$.

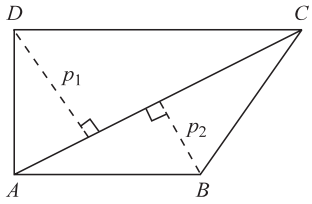
### 6. Quadrilateral

A closed figure bounded by four sides is called a *quadrilateral*.

It has four angles included in it.

The sum of these four angles is 360°.

i.e., $\angle A + \angle B + \angle C + \angle D = 360°$.




Area $(A)$ of a quadrilateral

$= \frac{1}{2} \times$ one diagonal $\times$ (sum of the perpendiculars to it from opposite vertices)

$= \frac{1}{2}d(p_1 + p_2)$

**Notes**

If the lengths of four sides and one of its diagonals are known, then,

$A$ = Area of $\Delta ADC$ + Area of $\Delta ABC$.

The special cases of quadrilateral are parallelogram, rectangle, square, rhombus, trapezium, etc., which are discussed below separately.

### 7. Parallelogram

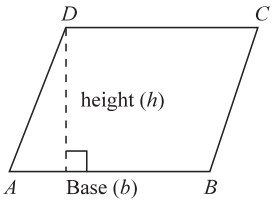
A quadrilateral in which opposite sides are equal and parallel is called a parallelogram.

The diagonals of a parallelogram bisect each other.

Area $(A)$ of a parallelogram

= base $\times$ altitude corresponding to the base

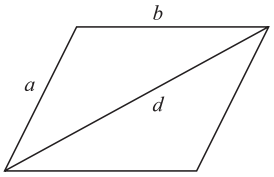
$= b \times h$




Area $(A)$ of a parallelogram

$= 2\sqrt{s(s-a)(s-b)(s-d)}$

where $a$ and $b$ are adjacent sides, $d$ is the length of the diagonal connecting the ends of the two sides and $s = \frac{a+b+d}{2}$.




In a parallelogram, the sum of the squares of the diagonals = 2 (the sum of the squares of the two adjacent sides),

i.e., $d_1^2 + d_2^2 = 2(a^2 + b^2)$.

Perimeter ($P$) of a parallelogram

$$= 2(a + b),$$

where $a$ and $b$ are adjacent sides of the parallelogram.

**Illustration 10:** One side of a parallelogram is 15 cm and the corresponding altitude is 5 cm. Find the area of the parallelogram.

**Solution:** Area of the parallelogram

$$= \text{base} \times \text{corresponding altitude}$$

$$= 15 \times 5 = 75 \text{ cm}^2.$$

**Illustration 11:** In a parallelogram, the lengths of the adjacent sides are 11 cm and 13 cm, respectively. If the length of one diagonal is 20 cm then, find the length of the other diagonal.

**Solution:** We have,

$$d_1^2 + d_2^2 = 2(a^2 + b^2)$$

$$\Rightarrow \quad (20)^2 + d_2^2 = 2(11^2 + 13^2)$$

$$\Rightarrow \quad d_2^2 = 2(121 + 169) - 400 = 180.$$

$$\therefore \quad d_2 = \sqrt{180} = 13.4 \text{ m (approx.)}$$

**Illustration 12:** Find the area of a quadrilateral of whose diagonal is 38 cm long and the lengths of perpendiculars from the other two vertices are 31 cm and 19 cm, respectively.

**Solution:** Area of the quadrilateral

$$= \frac{1}{2} d(p_1 + p_2)$$

$$= \frac{1}{2} \times 38 \times (31 + 19)$$

$$= 19 \times 50 = 950 \text{ cm}^2.$$

**Illustration 13:** Find the area of a parallelogram whose two adjacent sides are 130 m and 140 m and one of the diagonals is 150 m long.

**Solution:** Here, $a$ = 130 m, $b$ = 140 m and $d$ = 150 m

$$\therefore \quad s = \frac{a+b+d}{2} = \frac{130+140+150}{2} = \frac{420}{2} = 210 \text{ m}$$

$\therefore$ Area of the parallelogram

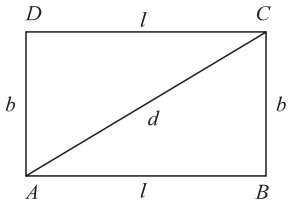
$$= 2\sqrt{s(s-a)(s-b)(s-a)}$$

$$= 2\sqrt{210(210-130)(210-140)(210-150)}$$

$$= 2\sqrt{210 \times 80 \times 70 \times 60} = 2 \times 8400 \text{ m}^2.$$

### 8. Rectangle

A *rectangle* is a quadrilateral with opposite sides equal and all the four angles equal to 90°.




The diagonals of a rectangle bisect each other and are equal.

(a) Area ($A$) of a rectangle = length × breadth

$$= l \times b$$

or,

Area of a rectangle = $(l \times \sqrt{d^2 - l^2})$,

if one side ($l$) and diagonal ($d$) are given.

or,

Area of a rectangle = $\left(\frac{p^2}{8} - \frac{d^2}{2}\right)$,

if perimeter ($P$) and diagonal ($d$) are given.

(b) Perimeter ($P$) of a rectangle

= 2(length + breadth)

= 2($l$ + $b$).

or,

Perimeter of a rectangle = $2(l + \sqrt{d^2 - l^2})$,

if one side ($l$) and diagonal ($d$) are given.

(c) Diagonal of a rectangle

$$= \sqrt{(\text{length})^2 + (\text{breadth})^2}$$

$$= \sqrt{l^2 + b^2}$$

(d) If area ($A$) and perimeter ($P$) of a rectangle are given, then,

length of a rectangle $= \left(\sqrt{\frac{P^2}{16} - A} + \frac{P}{4}\right)$

and,

breadth of a rectangle $= \left(\frac{P}{4} - \sqrt{\frac{P^2}{16} - A}\right)$

**Illustration 14:** Find the diagonal of a rectangle whose sides are 8 cm and 6 cm.

**Solution:** Diagonal of the rectangle

$= \sqrt{l^2 + b^2}$

$= \sqrt{8^2 + 6^2} = \sqrt{64 + 36}$

$= 10$ cm.

**Illustration 15:** Find the perimeter of a rectangle of length 12 m and breadth 6 m.

**Solution:** Perimeter of the rectangle

$= 2(l + b)$

$= 2(12 + 6) = 36$ m.

**Illustration 16:** Calculate the area of a rectangular field whose length is 12.5 cm and breadth is 8 cm.

**Solution:** Area of the rectangular field

$= l \times b$

$= 12.5 \times 8 = 100$ cm$^2$.

**Illustration 17:** Calculate the area of a rectangular field whose one side is 16 cm and the diagonal is 20 cm.

**Solution:** Area of the rectangular field

$= (l \times \sqrt{d^2 - l^2})$

$= (16 \times \sqrt{20^2 - 16^2})$

$= 16 \times 12 = 192$ cm$^2$.

**Illustration 18:** A rectangular carpet has an area of 120 m$^2$ and perimeter of 46 m. Find the length of its diagonal.

**Solution:** We have,

Area of rectangle $= \left(\frac{P^2}{8} - \frac{d^2}{2}\right)$

$\Rightarrow \quad 120 = \frac{46^2}{8} - \frac{d^2}{2}$

$\Rightarrow \quad 46^2 - 4d^2 = 120 \times 8$

$\Rightarrow \quad 4d^2 = 2116 - 960 = 1156$

$\therefore \quad d = \sqrt{289} = 17$ m.

**Illustration 19:** The perimeter of a rectangle is 82 cm and its area is 400 m$^2$. Find the length and breadth of the rectangle.

**Solution:** Length of the rectangle

$= \left(\sqrt{\frac{P^2}{16} - A} + \frac{P}{4}\right)$

$= \left(\sqrt{\frac{(82)^2}{16} - 400} + \frac{82}{4}\right)$

$= (4.5 + 20.5) = 25$ m.

Breadth of the rectangle

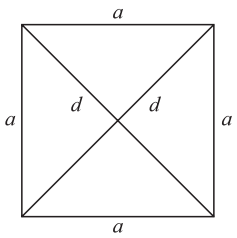
$= \left(\frac{P}{4} - \sqrt{\frac{P^2}{16} - A}\right)$

$= \left(\frac{82}{4} - \sqrt{\frac{(82)^2}{16} - 400}\right)$

$= (20.5 - 4.5) = 16$ m.

### 9. Square

A square is a quadrilateral with all sides equal and all the four angles equal to 90°.




The diagonals of a square are equal and bisect each other at 90°.

(a) Area ($A$) of a square

$= a^2$, i.e., (side)$^2$

$= \frac{d^2}{2}$, i.e., $\frac{(\text{diagonal})^2}{2}$

$= \frac{P^2}{16}$, i.e., $\frac{(\text{perimeter})^2}{16}$

(b) Perimeter ($P$) of a square

$= 4a$, i.e., $4 \times$ side

$= \sqrt{16 \times \text{area}}$

$= 2\sqrt{2}d$, i.e., $2\sqrt{2} \times$ diagonal

(c) Length ($d$) of the diagonal of a square

$= \sqrt{2a}$, i.e., $\sqrt{2} \times$ side

$= \sqrt{2 \times \text{area}}$

$= \dfrac{P}{2\sqrt{2}}$, i.e., $\dfrac{\text{Perimeter}}{2\sqrt{2}}$.

**Illustration 20:** If the area of a square field is 6050 $m^2$, find the length of its diagonal.

**Solution:** Length of the diagonal of the square field

$= \sqrt{2 \times \text{area}}$

$= \sqrt{2 \times 6050}$

$= \sqrt{12100}$, i.e., 110 m.

**Illustration 21:** Find the area of a square with perimeter 48 m.

**Solution:** Area of the square

$= \dfrac{(\text{Perimeter})^2}{16}$

$= \dfrac{(48)^2}{16} = \dfrac{48 \times 48}{16} = 3 \times 48 = 144 \text{ m}^2$.

**Illustration 22:** Find the diagonal of a square field whose side is 6 m.

**Solution:** Length of the diagonal

$= \sqrt{2} \times$ side

$= 6\sqrt{2}$ m.

**Illustration 23:** In order to fence a square, Ramesh fixed 36 poles. If the distance between the two poles is 6 m, then find the area of the square so formed.

**Solution:** Perimeter of the square

$= 36 \times 6 = 216$ m.

$\therefore$ Area of the square $= \left(\dfrac{\text{Perimeter}}{4}\right)^2$

$= \dfrac{16\sqrt{2}}{2\sqrt{2}} = 54 \times 54$

$= 2916 \text{ m}^2$.

**Illustration 24:** Perimeter of a square field is $16\sqrt{2}$ cm. Find the length of its diagonal.

**Solution:** We have,

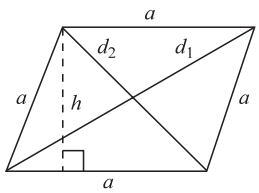
Perimeter of the square field $= 2\sqrt{2} \times$ diagonal

$\Rightarrow \quad 16\sqrt{2} = 2\sqrt{2} \times$ diagonal

$\therefore$ Length of the diagonal $= \dfrac{16\sqrt{2}}{2\sqrt{2}} = 8$ cm.

### 10. Rhombus

A *rhombus* is a quadrilateral whose all sides are equal.




The diagonals of a rhombus bisect each other at 90°.

(a) Area ($A$) of a rhombus

$= a \times h$, i.e., base $\times$ height

$= \dfrac{1}{2}d_1 \times d_2$, i.e., $\dfrac{1}{2} \times$ product of its diagonals

$= d_1 \times \sqrt{a^2 - \left(\dfrac{d_1}{2}\right)^2}$,

since, $d_2^2 = 4\left[a^2 - \left(\dfrac{d_1}{2}\right)^2\right]$

$= d_1 \times \sqrt{\left(\dfrac{\text{Perimeter}}{4}\right)^2 - \left(\dfrac{d_1}{2}\right)^2}$,

since, $d_2^2 = 4\left[\left(\dfrac{\text{Perimeter}}{4}\right)^2 - \left(\dfrac{d_1}{2}\right)^2\right]$

(b) Perimeter ($P$) of a rhombus

$= 4a$, i.e., $4 \times$ side

$= 2\sqrt{d_1^2 + d_2^2}$,

where $d_1$ and $d_2$ are two diagonals.

(c) Side ($a$) of a rhombus

$= \dfrac{1}{2}\sqrt{d_1^2 + d_2^2}$.

**Illustration 25:** The area of a rhombus is 156 $m^2$. If one of its diagonals is 13 $m^2$ then, find the length of the other diagonal.

**Solution:** Area of rhombus $= \frac{1}{2}(d_1 \times d_2)$

$\Rightarrow \quad 156 = \frac{1}{2}(13 \times d_2)$

$\Rightarrow \quad d_2 = \frac{2\times156}{13} = 24$ m.

**Illustration 26:** Find the area of a rhombus whose one side is 13 cm and one diagonal is 24 cm.

**Solution:** Area of rhombus $= d_1 \times \sqrt{a^2 - \left(\frac{d_1}{2}\right)^2}$

$= 24 \times \sqrt{(13)^2 - \left(\frac{24}{2}\right)^2}$

$= 24 \times \sqrt{169-144}$

$= 24 \times 5$

$= 120$ $cm^2$.

**Illustration 27:** If the perimeter of a rhombus is 73 cm and one of its diagonals is 27.5 cm then, find the other diagonal and the area of the rhombus.

**Solution:** One side of rhombus $(a) = \frac{73}{4} = 18.25$ cm.

$\therefore$ Other diagonal $(d_2) = 2 \times \sqrt{a^2 - \left(\frac{d_1}{2}\right)^2}$

$= 2 \times \sqrt{(18.25)^2 - \left(\frac{27.5}{2}\right)^2}$

$= 24$ cm.

$\therefore$ Area of rhombus $= \frac{1}{2}\times d_1 \times d_2$

$= \frac{1}{2}\times 24 \times 27.5$

$= 330$ $cm^2$.

**Illustration 28:** In a rhombus, the lengths of two diagonals are 18 m and 24 m. Find its perimeter.

**Solution:** Perimeter of the rhombus

$= 2 \times \sqrt{d_1^2 + d_2^2}$

$= 2 \times \sqrt{(18)^2 + (24)^2}$

$= 2 \times \sqrt{324+576}$

$= 2 \times \sqrt{900} = 60$ m.

**Illustration 29:** Find the side of a rhombus, one of whose diagonals measure 4 m and the other 3 m.

**Solution:** Side of the rhombus

$= \frac{1}{2}\times\sqrt{d_1^2 + d_2^2}$

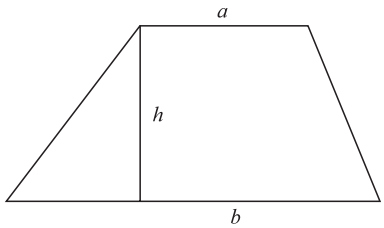
$= \frac{1}{2}\times\sqrt{(4)^2 + (3)^2}$

$= \frac{1}{2}\times 25$, i.e., 12.5 m.

## 11. Trapezium (Trapezoid)

A *trapezium* is a quadrilateral whose any two opposite sides are parallel.

Distance between the parallel sides of a trapezium is called its height.




(a) Area $(A)$ of a trapezium

$= \frac{1}{2} \times$ (sum of the parallel sides) $\times$ perpendicular distance between the parallel sides

i.e., $\frac{1}{2} \times (a + b) \times h$.

$= \frac{a+b}{l}\sqrt{s(s-l)(s-c)(s-d)}$,

where, $l = b - a$ if $b > a$

$= a - b$ if $a > b$

and, $s = \frac{c+d+l}{2}$

(b) Height $(h)$ of the trapezium

$= \frac{2}{l}\sqrt{s(s-l)(s-c)(s-d)}$

$= \left(\frac{2A}{a+b}\right)$

**Illustration 30:** Find the area of a trapezium having parallel sides 65 m and 44 m and distance between them being 20 m.

**Solution:** Area of the trapezium

$$= \frac{1}{2} \times (a + b) \times h$$

$$= \frac{1}{2} \times (65 + 44) \times 20$$

$$= 1100 \text{ cm}^2.$$

**Illustration 31:** The parallel sides of a trapezium are 24 m and 52 m. If its other two sides are 26 m and 30 m, then what is the area of the trapezium?

**Solution:** Area of the trapezium

$$= \frac{a+b}{l}\sqrt{s(s-l)(s-c)(s-d)}\,.$$

Here, $a = 24$, $b = 52$, $c = 26$, $d = 30$, $l = b - a = 28$,

$$s = \frac{c+d+l}{2} = \frac{26+30+28}{2} = 42.$$

$\therefore$ Area of the trapezium

$$= \frac{24+52}{28}\sqrt{42(42-28)(42-26)(42-30)}$$

$$= \frac{76}{28} \times \sqrt{42 \times 14 \times 16 \times 12}$$

$$= \frac{76 \times 336}{28} = 912 \text{ m}^2.$$

**Illustration 32:** The two parallel sides of a trapezium of area 180 cm$^2$ measure 28 cm and 12 cm. What is the height of the trapezium?

**Solution:** Height of the trapezium

$$= \left(\frac{2A}{a+b}\right) = \left(\frac{2 \times 180}{28+12}\right) = \frac{360}{40} = 9 \text{ cm}.$$

**12. Walls of a Room**

Area of four walls of a room

= 2(length + breadth) × height

**Illustration 33:** Find the cost of painting the walls of a room of 6 m long, 5 m broad and 4 m high at ₹7.50 per m$^2$.

**Solution:** Area of 4 walls of the room

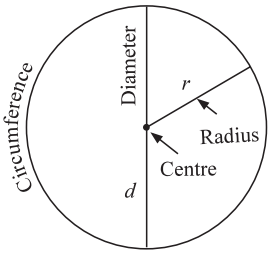
= 2(length + breadth) × height

$$= 2(6 + 5) \times 4 = 88 \text{ m}^2.$$

$\therefore$ Cost of painting = 88 × 7.50 = ₹660.

**12. Circle**

A *circle* is the path travelled by a point which moves in such a way that its distance from a fixed point remains constant.




The fixed point is known as *centre* and the fixed distance is called the *radius*.

(a) Circumference or perimeter of a circle

$$= 2\pi r = \pi d,$$

where $r$ is radius and $d$ is diameter of the circle

(b) Area of a circle

$= \pi r^2$, $r$ is radius

$= \frac{\pi d^2}{4}$, $d$ is diameter

$= \frac{c^2}{4\pi}$, $c$ is circumference

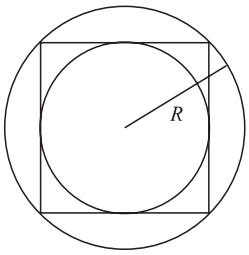
$= \frac{1}{2} \times$ circumference × radius

(c) Radius of a circle $= \sqrt{\frac{\text{Area}}{\pi}}$

$$= \frac{\text{Perimeter or circumference}}{2\pi}$$

(d) Ratio of the areas of the two circles is

$$= \frac{\text{Area of circle circumscribing the square}}{\text{Area of circle inscribed in the square}} = \frac{2}{1}.$$




(e) Ratio of the area of the two squares is

$$= \frac{\text{Area of square circumscribing the circle}}{\text{Area of square inscribed in the circle}} = \frac{2}{1}.$$

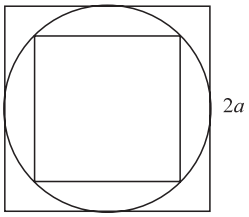



**Illustration 34:** What is the radius of a circular plot whose circumference is 176 m?

**Solution:** $r = \dfrac{\text{Circumference}}{2\pi}$

$= \dfrac{176}{2\times\frac{22}{7}} = \dfrac{176\times 7}{2\times 22} = 28$ m.

**Illustration 35:** A circular plot covers an area of 154 $m^2$. How much wire is required for fencing the plot?

**Solution:** Area of the plot $= \pi r^2 = 154$

i.e., $r^2 = 154 \times \dfrac{7}{22} = 49$

$\therefore \quad r = 7$ m

$\therefore$ Length of the wire $= 2\pi r = 2 \times \dfrac{22}{7}\times 7 = 44$ m.

**Illustration 36:** Find the length of a rope by which a buffalo must be tethered in order that she may be able to graze an area of 9856 $m^2$.

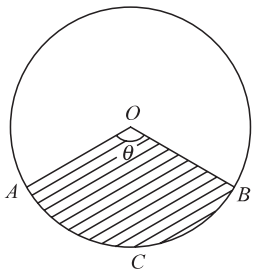
**Solution:** The required length of the rope

$= r = \sqrt{\dfrac{\text{Area}}{\pi}}$

$= \sqrt{\dfrac{9856\times 7}{22}} = \sqrt{3136} = 56$ m.

## Sector

A *sector* is a figure enclosed by two radii and an arc lying between them.




For sector $AOB$,

Arc $AB = \dfrac{2\pi r\theta}{360}$,

where $r$ = radius and $\angle AOB = \theta$

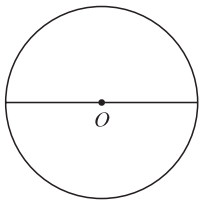
Area of the sector $ACBO$

$= \dfrac{1}{2} \times (\text{arc } AB) \times \text{radius}$

$= \dfrac{\pi(\text{radius})^2\theta}{360}$.

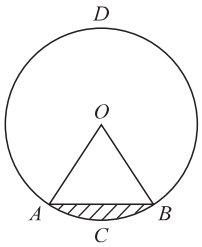
## Semi-Circle

A *semi-circle* is a figure enclosed by a diameter and the part of the circumference cut off by it.




## Segment

A *segment* of a circle is a figure enclosed by a chord and an arc which it cuts off.




**Notes**

Any chord of a circle which is not a diameter (such as AB), divides the circle into two segments, one is greater and the other is less than a semi-circle.

Area of segment $ACB$

= area of sector $ACBO$ – area of $\Delta OAB$

and area of segment $ADB$

= area of circle – area of segment $ACB$

**Illustration 37:** If a piece of wire 20 cm long is bent into an arc of a circle subtending an angle of 60° at the centre, find the radius of the circle.

**Solution:** Length of the arc $= \dfrac{2\pi r\theta}{360}$

$\Rightarrow \quad 20 = \dfrac{2\pi r\times 60}{360} \Rightarrow r = \dfrac{20\times 360}{60\times 2\times\pi} = \dfrac{60}{\pi}$ cm.

**Illustration 38:** Find the area of sector of a circle whose radius is 14 cm and the angle at the centre is 60°.

**Solution:** Area of thesector $= \frac{\pi(\text{radius})^2\theta}{360}$

$= \frac{22\times14\times14\times60}{7\times360}$

$= \frac{22\times2\times14}{6} = 102\frac{2}{3}$ cm$^2$.

**Illustration 39:** Find the area of sector of a circle whose radius is 10 cm and the length of the arc is 13 cm.

**Solution:** Area of the sector $= \frac{1}{2}\times$ (length of arc) $\times$ radius

$= \frac{1}{2} \times 13 \times 10$

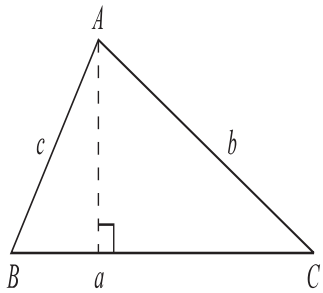
$= 65$ cm$^2$.

### Polygon

A *polygon* is a plane figure enclosed by four or more straight lines.

### Regular Polygon

If all the sides of a polygon are equal, it is called a *regular polygon.*




All the interior angles of a regular polygon are equal.

For a regular polygon:

Sum of exterior angles = $2\pi$

Sum of interior angles = $(n - 2)\pi$

Number of diagonals in a polygon = $\frac{n(n-3)}{2}$

Perimeter $(P) = n \times a$,

where, $n$ = number of sides

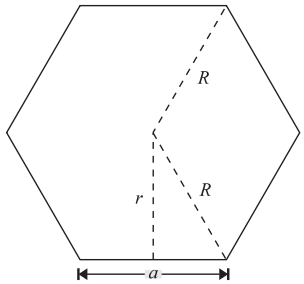
and, $a$ = length of each side

Each interior angle = $\frac{n-2}{n}\times\pi$

Each exterior angle = $\frac{2\pi}{n}$

Area = $\frac{1}{2} \times P \times r = \frac{1}{2} \times n \times a \times r$,

where, $r$ is radius of the circle drawn inside the polygon touching its sides.




$= \frac{1}{2} \times n \times a \times \sqrt{R^2 - \left(\frac{a}{2}\right)^2}$,

where, $R$ is radius of the circle drawn outside the polygon touching its sides.

$= \frac{na^2}{4}\cot\left(\frac{\pi}{n}\right)$.

Area of a regular hexagon = $\frac{3\sqrt{3}}{2}$ (side)$^2$

Area of a regular octagon = $2(\sqrt{2}+1)$ (side)$^2$.

**Illustration 40:** Find the side of a regular hexagon whose area is $48\sqrt{3}$ cm$^2$.

**Solution:** Area of a regular hexagon

$= \frac{3\sqrt{3}}{2} \times (\text{side})^2$

$\Rightarrow \quad 48\sqrt{3} = \frac{3\sqrt{3}}{2} \times (\text{side})^2$

$\Rightarrow \quad (\text{side})^2 = 32$

$\therefore$ Side of the hexagon = $4\sqrt{2}$ cm.

**Illustration 41:** Find the area of a regular octagon whose side measures $\sqrt{2}$ cm.

**Solution:** Area of regular octagon

$= 2(\sqrt{2}+1)\times(\text{side})^2$

$= 2(\sqrt{2}+1)\times(\sqrt{2})^2$

$= 4(\sqrt{2}+1)$ cm$^2$.

**Illustration 42:** Find the sum of interior angles of a regular polygon of 10 sides. Also, find the value of each interior angle.

**Solution:** Sum of interior angles $= (n - 2) \times \pi$

$= (10 - 2) \times \pi$

$= 8\pi.$

Also, value of each interior angle

$$= \left(\frac{n-2}{n}\right)\times\pi$$

$$= \left(\frac{10-2}{10}\right)\pi = \frac{4\pi}{5}.$$

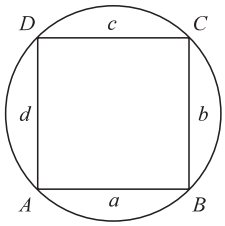
**Illustration 43:** Find the sum of all the exterior angles of a regular polygon of 12 sides. And, also find the value of each exterior angle.

**Solution:** Sum of exterior angles $= 2\pi$

Also, value of each exterior angle $= \frac{2\pi}{n} = \frac{2\pi}{12} = \frac{\pi}{6}.$

### Cyclic Quadrilateral

A quadrilateral whose vertices lie on the circumference of the circle is called a *cyclic quadrilateral*.




For a cyclic quadrilateral

- Area $= \sqrt{s(s-a)(s-b)(s-c)(s-d)}$,

  where,

  $$s = \frac{a+b+c+d}{2}$$
- $\angle A + \angle B + \angle C + \angle D = 2\pi$
- $\angle A + \angle C = \angle B + \angle D = \pi.$

## SHORT-CUT METHODS

**01** If the length and the breadth of a rectangle are increased by $x$% and $y$% respectively, then the area of the rectangle will increase by $\left(x + y + \frac{xy}{100}\right)$%

**Explanation:**

Area of the original rectangle is

$$A = l \times b.$$

Area of the new rectangle is

$$A' = l\left(\frac{100+x}{100}\right)\times b\left(\frac{100+y}{100}\right)$$

$$= lb\left(\frac{100+x}{100}\right)\left(\frac{100+y}{100}\right)$$

$$= A\left(\frac{100+x}{100}\right)\left(\frac{100+y}{100}\right)$$

$$\therefore \quad \frac{A'}{A} - 1 = \left(\frac{100+x}{100}\right)\left(\frac{100+y}{100}\right) - 1$$

or, $$\frac{A'-A}{A} = \frac{(100+x)(100+y)-(100)^2}{(100)^2}$$

or, $$\left(\frac{A'-A}{A}\right)\times 100$$

$$= \frac{(100)^2 + 100(x+y) + xy - (100)^2}{(100)}$$

$\therefore$ Percentage increase in area $= \frac{100(x+y)+xy}{100}$ %

or, $\left(x + y + \frac{xy}{100}\right)$%

**Notes**

If any of $x$ or $y$ decreases, we put negative sign.

**Illustration 1:** The length and the breadth of a rectangle are increased by 20% and 5%, respectively. Find the percentage increase in its area.

**Solution:** Percentage increase in the area of rectangle

$$= \left(x + y + \frac{xy}{100}\right)\%$$

$$= \left(20 + 5 + \frac{20\times 5}{100}\right)\% = 20\%$$

**02** If the length of a rectangle is increased by $x$%, then its breadth will have to be decreased by $\left(\frac{100x}{100+x}\right)$% in order to maintain the same area of rectangle.

**Explanation:**

Percentage increase in area of rectangle

$$= \left(x + y + \frac{xy}{100}\right)\%$$

or, $$0 = \left(x + y + \frac{xy}{100}\right)$$

or, $$-x = y\left(1 + \frac{x}{100}\right) \quad \text{or,} \quad y = -\left(\frac{100x}{100 + x}\right)$$

–ve sign indicates decrease.

Therefore, breadth must be decreased by $\left(\frac{100x}{100 + x}\right)\%$ in order to maintain the same area.

**Illustration 2:** The length of a rectangle is increased by 25%. By what per cent should its breadth be decreased so as to maintain the same area?

**Solution:** The breadth must be decreased by

$$= \left(\frac{100x}{100 + x}\right)\% = \left(\frac{100 \times 25}{100 + 25}\right)\%, \text{ i.e., } 20\%$$

**03** If each of the defining dimensions or sides of any two-dimensional figure (triangle, rectangle, square, circle, quadrilateral, pentagon, hexagon, etc.) is changed by $x$%, its area changes by $x\left(2 + \frac{x}{100}\right)\%$

**Illustration 3:** If the radius of a circle is decreased by 10%, what is the percentage decrease in its area?

**Solution:** Here, $x = -10$ (–ve sign indicates decrease)

$\therefore$ Percentage change in area $= x\left(2 + \frac{x}{100}\right)\%$

$$= -10\left(2 - \frac{10}{100}\right)\%$$

$$= (-10)\left(\frac{19}{10}\right)\% = -19\%$$

$\therefore$ Area of the circle decreases by 19%

**04** If all the sides of a quadrilateral are increased (or decreased) by $x$%, its diagonals also increase (or decrease) by $x$%

**Illustration 4:** The length and the two diagonals of a rectangle are decreased by 5% each. What is the percentage decrease in its breadth?

**Solution:** Since the length and the two diagonals decreased by 5% each, the breadth also must decrease by 5%.

**05** If each of the defining dimensions or sides of any two-dimensional figures are increased (or decreased) by $x$%, its perimeter also increases (or decreases) by $x$%

**Illustration 5:** If all the sides and diagonals of a square are increased by 8% each, then find the percentage increase in its perimeter.

**Solution:** The perimeter also increases by 8%

**06** If the ratio of the areas of two squares be $a$:$b$, then the ratio of their sides, ratio of their perimeters and the ratio of their diagonals, each will be in the ratio $\sqrt{a} : \sqrt{b}$.

**Illustration 6:** Ratio of the areas of two squares is 16:9. Find the ratio of their diagonals.

**Solution:** The ratio of their diagonals

$$= \sqrt{a} : \sqrt{b}$$
$$= \sqrt{16} : \sqrt{9}, \text{ i.e., } 4:3.$$

**07** If the diagonal of a square increases by $x$ times, then the area of the square becomes $x^2$ times.

**Illustration 7:** The diagonal of a square is doubled. How many times will the area of the new square become?

**Solution:** The area of the new square will become $x^2$ times, i.e., $(2)^2 = 4$ times.

**08** Standard Properties of Diagonals of Quadrilaterals

| *Quadrilateral* | *Meet at right angles* | *Bisect each other* | *Equal to each other* | *Bisect angle at vertex* |
|---|---|---|---|---|
| Square | √ | √ | √ | √ |
| Rectangle | × | √ | √ | × |
| Parallelogram | × | √ | × | × |
| Rhombus | √ | √ | × | √ |
| Trapezium | × | × | may or may not be | × |

**09** Carpeting the Floor of a Room

If the length and the breadth of a room are $l$ and $b$, respectively, and a carpet of width $w$ is used to cover the floor, then the required length of the carpet

$$= \frac{l \times b}{w}.$$

**Illustration 8:** How many metres of a carpet of 12 cm wide will be required to cover the floor of a room which is 600 cm long and 420 cm broad? Also, calculate the amount required in carpeting the floor if the cost of carpet is ₹15 per metre.

**Solution:** Length of the carpet

$$= \frac{l \times b}{w}$$

$$= \frac{600 \times 420}{12}$$

$$= 21000 \text{ cm, i.e., } 210 \text{ m.}$$

The amount required for carpeting the floor

$$= 15 \times 210 = ₹3150.$$

**10** Number of Square Tiles Required for Flooring

If the length and the breadth of a room are $l$ and $b$ respectively, then the least number of square tiles required to cover the floor

$$= \frac{l \times b}{\text{H.C.F. } (l, b)}$$

Also, the size of the largest tile so that the tiles exactly fit

$$= \text{H.C.F. } (l, b).$$

**Illustration 9:** A hall of length 24 cm and breadth 20 m is to be paved with equal square tiles. What will be the size of the largest tile so that the tiles exactly fit and also find the number of tiles required.

**Solution:** Size of the largest possible square tile

$$= \text{H.C.F. } (l, b)$$

$$= \text{H.C.F. } (24, 20) = 4 \text{ m}$$

Number of tiles required

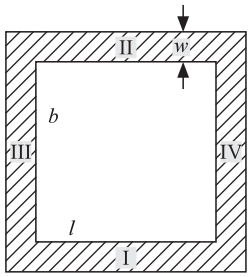
$$= \frac{l \times b}{\text{H.C.F. } (l, b)}$$

$$= \frac{24 \times 20}{4} = 120 \text{ tiles.}$$

**11** Path around a Rectangular Space

(a) A rectangular garden $l$ m long and $b$ m broad is sorrounded by a path of $w$ m wide. The area of the path is given by

$$= 2w(l + b + 2w) \text{ m}^2.$$




**Explanation:**

Area of part I = Area of part II

$$= (l + 2w)w \text{ m}^2$$

Area of part III = Area of part IV

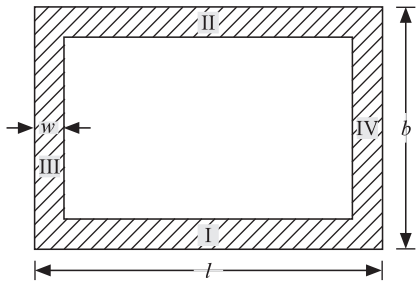
$$= bw \text{ m}^2.$$

$\therefore$ Total area of the path $= 2[(l + 2w)w + bw]$

$$= 2w(l + b + 2w) \text{ m}^2.$$

(b) A rectangular garden $l$ m long and $b$ m broad is sorrounded by a path of $w$ m wide constructed inside it along its boundary. The area of the path is given by

$$= 2w(l + b - 2w) \text{ m}^2.$$




**Explanation:**

Area of part I = Area of part II

$$= lw \text{ m}^2$$

Area of part III = Area of part IV

$$= (b - 2w)w \text{ m}^2$$

$\therefore$ Total area of the path

$$= 2[lw + (b - 2w)w]$$

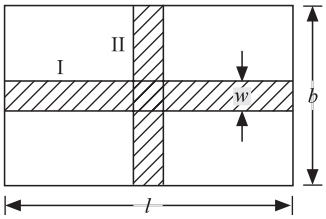
$$= 2w(l + b - 2w) \text{ m}^2.$$

(c) A rectangular park is $l$ m long and $b$ m broad. Two paths inside the park of $w$ m wide and they are perpendicular to each other. The area of the paths

$$= w(l + b - w) \text{ m}^2$$

Also, the area of the park minus the paths

$$= (l - w)(b - w) \text{ m}^2$$




**Explanation:**

Total area of the path

= Area of path I + Area of path II − Area of common central part

$= lw + bw - w^2$

$= w(l + b - w)$ m$^2$.

$\therefore$ Area of the park minus the paths

$= [lb - w(l + b - w)]$

$= lb - lw - w(b - w)$

$= l(b - w) - w(b - w)$

$= (l - w)\ (b - w)$ m$^2$.

**Notes**

1. Clearly, from the figure, the area of the paths does not change on shifting their location as long as they are perpendicular to each other.
2. For a square park, take $l = b$ in all the results derived above.

**Illustration 10:** A rectangular park 18 m × 12 m, is sorrounded by a path of 4 m wide. Find the area of the path.

**Solution:** The area of the path

$= 2w(l + b + 2w)$

$= 2 \times 4(18 + 12 + 2 \times 4) = 304$ m$^2$.

**Illustration 11:** A park is square in shape with side 18 m. Find the area of the pavement of 3 m wide to be laid all around it on its inside.

**Solution:** Area of the pavement

$= 2w(l + b - 2w)$

$= 2 \times 3(18 + 18 - 2 \times 3)$ (Here, $l = b = 18$)

$= 180$ m$^2$.

**Illustration 12:** A playground measures 27 m × 13 m. From the centre of each side a path 2 m wide goes across to the centre of the opposite side. Calculate the area of the path and the cost of constructing it at ₹4 per m$^2$.

**Solution:** Area of the path

$= w(l + b - w) = 2(27 + 13 - 2)$

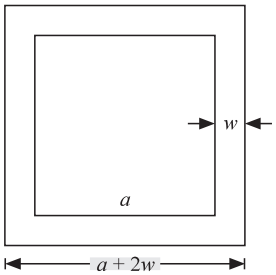
$= 76$ m$^2$.

$\therefore$ Cost $= 4 \times 76 =$ ₹304.

## 12 Square Room Surrounded by a Verandah

(a) A square room of side $a$ is sorrounded by a verandah of width $w$ on the outside of the square room. If the area of the verandah is $A$, then the area of the room is given by

$$\left(\frac{A-4w^2}{4w}\right)^2.$$




**Explanation:**

Area of the room $= a^2$.

Area of the (room + verandah) $= (a + 2w)^2$.

$\therefore$ Area ($A$) of the verandah $= (a + 2w)^2 - a^2$

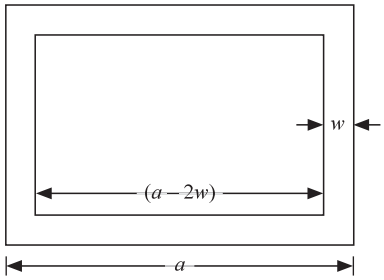
$= (4aw + 4w^2)$

or, $a = \dfrac{A-4w^2}{4w}$

$\therefore$ Area of the room $= a^2 = \left(\dfrac{A-4w^2}{4w}\right)^2$.

(b) A square room of side $a$ is surrounded by a verandah of width $w$ on its inside. If the area of the verandah is A, then the area of the room is given by

$$\left(\frac{A+4w^2}{4w}\right)^2.$$




**Explanation:**

Area ($A$) of the verandah $= a^2 - (a - 2w)^2$

$= 4aw - 4w^2$

$= 4w(a - w)$

or, $a = \frac{A}{4w} + w = \frac{A+4w^2}{4w}$

$\therefore$ Area of the room $= a^2 = \left(\frac{A+4w^2}{4w}\right)^2$.

**Illustration 13:** A square field is surrounded by a path of 2 m wide on its outside. The area of the path is 72 $m^2$. What is the area of the field?

**Solution:** Area of the field

$$= \left(\frac{A-4w^2}{4w}\right)^2$$

$$= \left(\frac{72-4\times 2^2}{4\times 2}\right)^2 = 49 \text{ m}^2.$$

**Illustration 14:** A square room has a verandah of area 24 $m^2$ and width 1 m all around it on its inside. Find the area of the room.

**Solution:** Area of the room

$$= \left(\frac{A+4w^2}{4w}\right)^2$$

$$= \left(\frac{24+4\times 1^2}{4\times 1}\right)^2 = 49 \text{ m}^2.$$

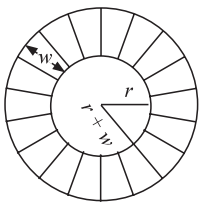
**13** (a)A circular ground of radius $r$ has a pathway of width $w$ around it on its outside. The area of the circular pathway is given by
$= \pi w(2r + w)$.

**Explanation:**

Area of the circular ground $= \pi r^2$

Area of the circular ground + pathway

$$= \pi(r + w)^2 = \pi r^2 + 2\pi rw + \pi w^2.$$




$\therefore$ Area of circular the pathway

$$= (\pi r^2 + 2\pi rw + \pi w^2) - \pi r^2$$

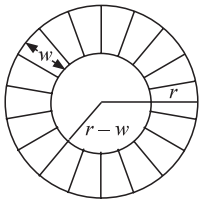
$$= \pi w(2r + w).$$

(b) A circular ground of radius $r$ has a pathway of width $w$ around it on its inside. The area of the circular pathway is given by

$$= \pi w(2r - w).$$

**Explanation:**

Area of circular the ground $= \pi r^2$




Area of the circular ground – pathway

$$= \pi (r - w)^2$$

$$= \pi r^2 - 2\pi rw + \pi w^2$$

$\therefore$ Area of the circular pathway

$$= \pi r^2 - (\pi r^2 - 2\pi rw + \pi w^2)$$

$$= \pi w(2r - w).$$

**Illustration 15:** A circular ground of radius 16 m has a path of width 2.8 m around it on its outside. Find the area of the path.

**Solution:** The area of the circular path

$$= \pi\, w(2r + w)$$

$$= \frac{22}{7} \times 2.8 \times (2 \times 16 + 2.8)$$

$$= 8.8 \times (32 + 2.8) = 306.2 \text{ m}^2.$$

**Illustration 16:** A circular park of radius 22 m has a path of width 1.4 m around it on its inside. Find the area of the path.

**Solution:** The area of the circular path

$$= \pi\, w(2r - w)$$

$$= \frac{22}{7} \times 1.4 \times (2 \times 22 - 1.4)$$

$$= 4.4 \times (44 - 1.4) = 187.45 \text{ m}^2.$$

**14** If the area of a square is $a$ $cm^2$, then the area of the circle formed by the same perimeter is $\left(\frac{4a}{\pi}\right)$ $cm^2$.

**Explanation:**

Area of the square $= a$.

$\therefore$ Side of the square $= \sqrt{\text{Area}} = \sqrt{a}$ .

$\therefore$ Perimeter of the square $= 4\sqrt{a}$ .

Given, Circumference of the circle = Perimeter of the square

$\Rightarrow$ $2\pi r = 4\sqrt{a}$

$\therefore$ Radius of the circle $(r) = \dfrac{4\sqrt{a}}{2\pi} = \dfrac{2\sqrt{a}}{\pi}$

$\therefore$ Area of the circle $= \pi r^2 = \pi \left(\dfrac{2\sqrt{a}}{\pi}\right)^2 = \dfrac{4a}{\pi}$ cm$^2$.

**Illustration 17:** If the area of a square is 33 cm$^2$, then find the area of the circle formed by the same perimeter.

**Solution:** Required area of the circle

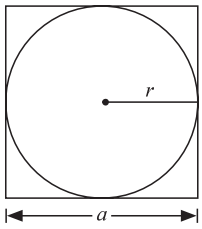
$$= \frac{4a}{\pi}$$

$$= \frac{4\times 33}{\pi} = \frac{4\times 33\times 7}{22} = 42 \text{ cm}^2.$$

**15** The area of the largest circle that can be inscribed in a square of side $a$ is $\dfrac{\pi a^2}{4}$.

**Explanation**

Clearly, from the figure, the diameter of the inscribed circle equals the side of the square i.e., $D = a$.




Area of the circle $= \dfrac{\pi D^2}{4}$

$\therefore$ Area of the inscribed circle $= \dfrac{\pi a^2}{4}$.

**Illustration 18:** Find the area of largest circle inscribed in a square of side 112 cm.

**Solution:** The area of the largest circle

$$= \frac{\pi a^2}{4}$$

$$= \frac{22\times 112\times 112}{7\times 4} = 9856 \text{ cm}^2.$$

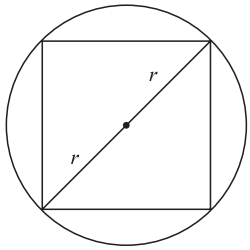
**16** Area of a square inscribed in a circle of radius $r$ is $2r^2$ and the side of the square is $\sqrt{2}\,r$.

**Explanation:**

Clearly, from the figure, diagonal of the inscribed square is equal to the diameter of the circle, i.e., $2r$.

$\therefore$ Area of square = (diagonal)$^2$

$$= \frac{1}{2}(2r)^2 = 2r^2.$$




Also, side of the square $= \sqrt{\text{Area}} = \sqrt{2r^2} = \sqrt{2}r$.

**Illustration 19:** Find the side of the square inscribed in a circle whose circumference is 308 cm.

**Solution:** Circumference of the circle $(2\pi r) = 308$

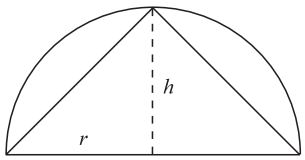
$\Rightarrow r = \dfrac{308}{2\pi} = \dfrac{308\times 7}{2\times 22} = 49$ cm.

$\therefore$ Side of the inscribed square $= \sqrt{2}\,r = 49\sqrt{2}$ cm.

**17** The area of the largest triangle inscribed in a semi-circle of radius $r$ is $r^2$.

**Explanation:**

Clearly, from the figure, the largest triangle inscribed in a semi-circle is an isosceles triangle with diameter as its base and radius as its height.




Area of the triangle $= \dfrac{1}{2} \times$ base $\times$ height

$$= \frac{1}{2} \times 2r \times r = r^2.$$

**18** The number of revolutions made by a circular wheel of radius $r$ in travelling a distance $d$ is given by

$$= \left(\frac{d}{2\pi r}\right).$$

**Explanation:**
Circumference of the wheel = $2\pi r$

In travelling a distance $2\pi r$, the wheel makes 1 revolution.

$\therefore$ In travelling a distance $d$, the wheel makes $\frac{22}{7}$ revolutions.

**Illustration 20:** The diameter of a wheel is 2 cm. If it rolls forward covering 10 revolutions, find the distance travelled by it.

**Solution:** Radius of the wheel = 1 cm.

The distance travelled by the wheel in 10 revolutions

$= 10 \times 2\pi r$

$= 10 \times 2 \times \frac{22}{7} \times 1 = 62.8$ cm.

## EXERCISE-I

**1.** The area of a triangle whose sides are 15 m, 16 m and 17 m is:
(a) $24\sqrt{4}$ m$^2$ (b) $24\sqrt{3}$ m$^2$
(c) $24\sqrt{21}$ m$^2$ (d) None of these

**2.** The area of a right-angled triangle with base 6 m and hypotenuse 6.5 m is:
(a) 7.5 m$^2$ (b) 9.5 m$^2$
(c) 8.5 m$^2$ (d) None of these

**3.** The length of each side of a triangle is 12 cm. The height of the triangle is:
(a) $3\sqrt{2}$ m (b) $6\sqrt{3}$ m
(c) $6\sqrt{2}$ (d) None of these

**4.** The area of a triangular lawn is 1600 m$^2$. If one side is 64 m long and the other two sides are equal in length, the length of each equal side is:
(a) 60.37 m (b) 59.36 m
(c) 60.36 cm (d) None of these

**5.** Three sides of a triangular field are 20 m, 21 m and 29 m long, respectively. The area of the field is:
(a) 215 m$^2$ (b) 230 m$^2$
(c) 210 m$^2$ (d) None of these

**6.** The hypotenuse and the semi-perimeter of a right triangle are 20 cm and 24 cm, respectively. The other two sides of the triangle are:
(a) 16 cm, 12 cm
(b) 20 cm, 12 cm
(c) 20 cm, 16 cm
(d) None of these

**7.** The sides of a triangle are in the ratio 3:4:5. If its perimeter is 36 cm, then the area of the triangle is:
(a) 57 m$^2$ (b) 54 m$^2$
(c) 56.5 m$^2$ (d) None of these

**8.** For a triangle whose sides are 50 m, 78 m and 112 m, respectively, then the length of the perpendicular from the opposite angle on the side 112 m is:
(a) 45 m (b) 35 m
(c) 30 m (d) None of these

**9.** A ladder is resting with one end in contact with the top of a wall of height 12 m and the other end on the ground is at a distance 5 m from the wall. The length of the ladder is:
(a) 13 m (b) 17 m
(c) 16 m (d) None of these

**10.** A ladder is placed so as to reach a window 63 cm high. The ladder is then turned over to the opposite side of the street and is found to reach a point 56 cm high. If the ladder is 65 cm long, then the width of the street is:
(a) 59 cm (b) 39 cm
(c) 49 cm (d) None of these

**11.** If the area of a triangle with base $x$ is equal to the area of a square with side $x$, then the altitude of the triangle is:
(a) $\frac{x}{2}$ (b) $x$
(c) $2x$ (d) $3x$

**12.** If the area of a triangle is 150 m$^2$ and the ratio of the base and the height is 3:4, then find its height.
(a) 25 m (b) 35 m
(c) 20 m (d) None of these

**13.** The base of a triangular field is three times its height. If the cost of cultivating the field at ₹36.72 per hectare is ₹495.72, then find its base and height.
(a) 950 m, 350 m (b) 800 m, 500 m
(c) 900 m, 300 m (d) None of these

**14.** If the sides of a triangle are doubled, its area:
(a) remains same (b) is doubled
(c) becomes 4 times (d) Can't say

**15.** Two sides of a triangular field are 85 m and 154 m, respectively, and its perimeter is 324 m. The cost of ploughing the field at the rate of ₹10 per $m^2$ is:
(a) ₹27720 (b) ₹37620
(c) ₹26750 (d) None of these

**16.** The area of an equilateral triangle, each of whose sides measures $2\sqrt{3}$ cm is:
(a) $5\sqrt{3}$ $m^2$ (b) $4\sqrt{3}$ $m^2$
(c) $3\sqrt{3}$ $cm^2$ (d) None of these

**17.** If the height of an equilateral triangle is $2\sqrt{3}$ cm, then the length of its side is:
(a) 4 cm (b) 6 cm
(c) 5 cm (d) None of these

**18.** If the perimeter of an equilateral triangle is 12 m, then find its area.
(a) $3\sqrt{4}$ m (b) $4\sqrt{3}$ m
(c) $5\sqrt{3}$ m (d) None of these

**19.** The height of an equilateral triangle whose perimeter is 24 cm, is:
(a) $4\sqrt{3}$ cm (b) $3\sqrt{4}$ cm
(c) $5\sqrt{3}$ cm (d) None of these

**20.** The perimeter of a right angled triangle is 90 cm and its hypotenuse is 39 cm. Find its other sides.
(a) 30 cm, 10 cm (b) 36 cm, 15 cm
(c) 48 cm, 20 cm (d) None of these

**21.** The perimeter of an isosceles triangle is 306 m and each of the equal sides is $\frac{5}{8}$ of the base. Find the area.
(a) 3648 cm (b) 3468 $m^2$
(c) 3846 cm (d) None of these

**22.** Find the area of an isosceles right-angled triangle whose hypotenuse is 8 cm.
(a) 32 $m^2$ (b) 24 $cm^2$
(c) 16 $cm^2$ (d) None of these

**23.** The perimeter of an isosceles triangle is equal to 14 cm; the lateral side and the base is in the ratio 5:4. The area of the triangle is:
(a) $3\sqrt{21}$ $cm^2$ (b) $2\sqrt{21}$ $cm^2$
(c) $4\sqrt{21}$ $cm^2$ (d) None of these

**24.** If all the sides of a triangle are increased by 200%, then the area of the triangle will increase by:
(a) 400% (b) 600%
(c) 800% (d) None of these

**25.** A plot of land is in the shape of a right-angled isosceles triangle. The length of the hypotenuse is $50\sqrt{2}$ m. The cost of fencing is ₹3 per metre. The total cost of fencing the plot will be:
(a) Less than ₹300
(b) Less than ₹500
(c) More than ₹500
(d) None of these

**26.** If the area of an equilateral triangle is equal to the area of an isosceles triangle whose base and equal sides are 16 cm and 10 cm respectively, then the side of the equilateral triangle is:
(a) 10.5 cm (b) 9.5 cm
(c) 12.5 cm (d) None of these

**27.** If the perimeter of a right-angled isosceles triangle is $4\sqrt{2} + 4$ m, then the hypotenuse is:
(a) 8 m (b) 6 m
(c) 4 m (d) None of these

**28.** The two adjacent sides of a parallelogram are 60 m and 40 m and one of the diagonals is 80 m long. The area of the parallelogram is:
(a) $600\sqrt{15}$ $m^2$ (b) $800\sqrt{25}$ $m^2$
(c) $700\sqrt{15}$ $m^2$ (d) None of these

**29.** One side of the parallelogram is 14 cm. Its distance from the opposite side is 16 cm. The area of the parallelogram is:
(a) 234 $cm^2$ (b) 324 $cm^2$
(c) 224 $cm^2$ (d) None of these

**30.** A field is in the shape of a parallelogram. Its adjacent sides and one diagonal are 65 m, 119 m and 156 m, respectively. Find the cost of gravelling it at the rate of ₹10 per $m^2$.
(a) ₹81400 (b) ₹71400
(c) ₹91400 (d) None of these

**31.** One side of a parallelogram is 10 m and the corresponding altitude is 7 m. The area of the parallelogram is:
(a) 70 $m^2$ (b) 60 $m^2$
(c) 80 $m^2$ (d) None of these

**32.** The adjacent sides of a parallelogram are 8 m and 5 m. The distance between the longer sides is 4 m. The distance between the shorter sides is:
(a) 4.6 m (b) 6.4 m
(c) 8.6 m (d) None of these

**33.** The area of a quadrilateral is 420 $m^2$ and the perpendiculars drawn to one diagonal from the

opposite vertices are 18 m and 12 m. Then, the length of the diagonal is:

(a) 32 m (b) 24 m
(c) 28 m (d) None of these

**34.** The area of a parallelogram is 72 $cm^2$ and its altitude is twice the corresponding base. The length of the base is:

(a) 6 cm (b) 8 cm
(c) 4 cm (d) None of these

**35.** The area of a parallelogram is 240 $cm^2$ and its height is 12 cm. The base of the parallelogram is:

(a) 24 cm (b) 20 cm
(c) 28 cm (d) None of these

**36.** In a quadrilateral *ABCD*, the sides, *AB*, *BC*, *CD*, *DA* measure 20 m, 13 m, 17 m and 10 m, respectively and the diagonal *AC* is 21 m. The area of the quadrilateral is:

(a) 210 $m^2$ (b) 220 $m^2$
(c) 240 $m^2$ (d) None of these

**37.** If the two diagonals of a parallelogram are 72 cm and 30 cm respectively, then find its perimeter.

(a) 156 cm (b) 164 cm
(c) 172 cm (d) None of these

**38.** If the base of a parallelogram is $(x + 4)$, altitude to the base is $(x - 3)$ and the area is $(x^2 - 4)$, then the actual area is equal to:

(a) 64 sq units (b) 48 sq units
(c) 60 sq units (d) None of these

**39.** In a parallelogram, the lengths of adjacent sides are 12 cm and 14 cm, respectively. If the length of one diagonal is 16 cm, then find the length of the other diagonal.

(a) 24.8 cm (b) 20.6 cm
(c) 22.4 cm (d) None of these

**40.** Find the perimeter of a circular plot which occupies an area of 154 $m^2$.

(a) 54 m (b) 44 m
(c) 22 m (d) 11 m

**41.** The perimeter of a circle is equal to that of a square. Compare their areas.

(a) 14:11 (b) 25:12
(c) 24:7 (d) 22:7

**42.** The length of a rectangle is thrice its breadth and its perimeter is 96 m. The area of the rectangle is:

(a) 288 $m^2$ (b) 442 $m^2$
(c) 438 $m^2$ (d) 432 $m^2$

**43.** A cow is tied by a rope at the corner of a rectangular field. If the length of the rope is 14 m, then the area of the field which the cow could graze is:

(a) 77 $m^2$ (b) 308 $m^2$
(c) 23 $m^2$ (d) 154 $m^2$

**44.** The wheel of a scooter has diameter 70 cm. How many revolutions per minute must the wheel make so that the speed of the scooter is kept 66 Km/h?

(a) 400 (b) 600
(c) 500 (d) 800

**45.** 2 small circular parks of diameters 16 m, 12 m are to be replaced by a bigger circular park. What would be the radius of this new park, if the new park occupies same space as the two small parks?

(a) 10 (b) 15
(c) 20 (d) 25

**46.** A rectangular park is 65 m long and 50 m wide. 2 cross paths each 2 m wide are to be constructed parallel to the sides. If these paths pass through the centre of the rectangle and cost of construction is ₹17.25 per $m^2$, then find the total cost involved in the construction.

(a) ₹2265.59 (b) ₹1772.45
(c) ₹3898.50 (d) ₹8452.32

**47.** The area of a trapezium is 2500 $m^2$. One of its parallel sides is 75 m. If the distance between the two parallel sides is 40 m, then find the length of the other parallel side.

(a) 20 m (b) 30 m
(c) 40 m (d) 50 m

**48.** The length of a rectangle exceeds its breadth by 3 cm. If the numerical values of the area and the perimeter of the rectangle are equal, then the breadth is:

(a) 1 cm (b) 2 cm
(c) 3 cm (d) 3.5 cm

**49.** If the ratio of the areas of two squares is 9:1, then the ratio of their perimeters is:

(a) 9:1 (b) 3:4
(c) 3:1 (d) 1:3

**50.** A square field with side 30 m is surrounded by a path of uniform width. If the area of the path is 256 $m^2$, then its width is:

(a) 16 m (b) 14 m
(c) 4 m (d) 2 m